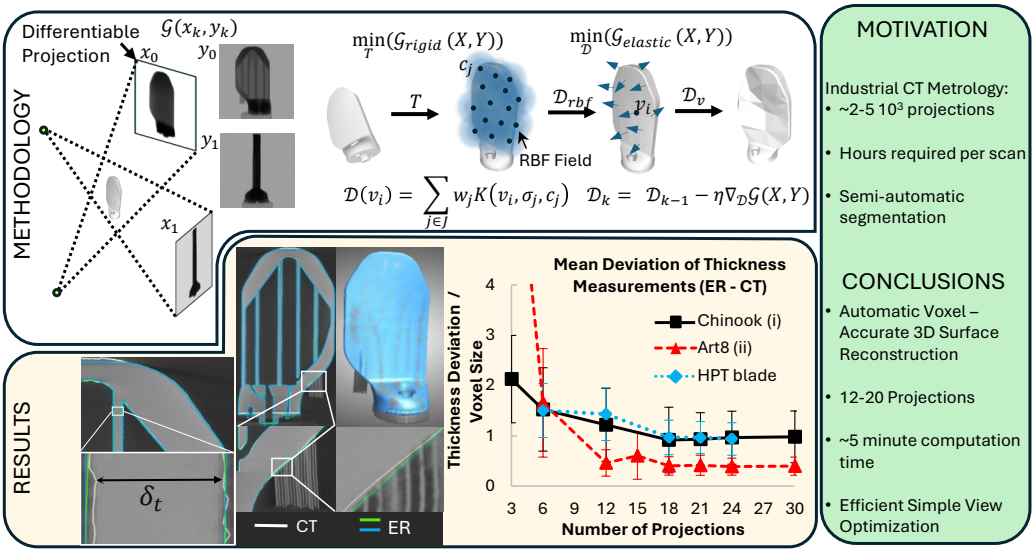
Graphical Abstract

CADER: CAD-constrained Elastic Registration for Voxel-Accurate Sparse-View 3D Surface Reconstruction in Minutes

Hristo Valtchanov, Weiqiang Liu, Sadaival Singh, Josselin Paccoud, Ammar Al Sheghri, Julien Dompierre, Catherine Desrosiers, Nicolas Piché, Vladimir Brailovski, François Guibault



MOTIVATION

Industrial CT Metrology:

- ~2-5 10^3 projections
- Hours required per scan
- Semi-automatic segmentation

CONCLUSIONS

- Automatic Voxel – Accurate 3D Surface Reconstruction
- 12-20 Projections
- ~5 minute computation time
- Efficient Simple View Optimization

Highlights

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- First differentiable-rendering-based elastic 3D–2D registration framework for industrial CT metrology that can directly deform a mesh to match CT images with voxel-level accuracy on par with full-view microCT.
- A hierarchical multi-resolution deformation model (multi-scale RBF fitting + constrained texture refinement) recovers surface detail while suppressing sparse-view artifacts.
- The proposed pipeline eliminates volumetric reconstruction and segmentation, yielding a direct surface output and a hundred-fold reduction in scan time and overall processing throughput.
- A view-optimization strategy is proposed that further improves accuracy and reduces the number of required images for complex geometries.
- The method enables high-throughput fully automated inspection using CT metrology on a per-part basis.
- Subsampling can retain metrological information, "sub-voxel" accuracy must be evaluated relative to the original unbinned voxel size.

CADER: CAD-constrained Elastic Registration for Voxel-Accurate Sparse-View 3D Surface Reconstruction in Minutes

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Abstract

Computed tomography (CT) is a standard for high-precision metrology in industrial non-destructive testing (NDT), enabling sub-voxel reconstruction of internal geometries in complex aerospace and biomedical components. However, attaining such accuracy typically requires thousands of X-ray projections and long acquisition and reconstruction times, particularly for high-attenuation materials such as nickel alloys. This work presents a computationally efficient alternative to volumetric CT reconstruction based on projection-space 3D–2D elastic registration of a prior CAD model to sparse CT projections. The voxel-wise reconstruction and segmentation steps are replaced by a hierarchical surface deformation model that morphs a CAD mesh using multi-scale radial basis functions (RBFs) followed by constrained vertex-level refinement (texture fitting). A differentiable X-ray renderer based on soft rasterization simulates forward projections and provides analytical gradients of mutual-information and edge-based objectives for gradient-based optimization with stochastic batching and sub-image sampling. A geometry-driven view-optimization heuristic further improves sparse-view performance using only the rigidly registered CAD input. Across

representative industrial datasets, measurement accuracy is equivalent to CT reconstruction and segmentation using filtered-back projection using between 12-20 projections. Wall thickness measurements relative to micro CT segmentation (filtered back projection) deviate by less than 1 voxel for simple parts, though both methods measure thickness within one voxel compared to coordinate-measuring-machine (CMM) measurements. For complex industrial components such as a turbine blade, mean wall thickness deviation is $\tilde{1}$ voxel vs conventional microCT segmentation. The proposed framework reduces image acquisition time to as little as 22 s with total processing time being less than five minutes (a hundred-fold improvement relative to conventional microCT for the same accuracy), establishing a realistic pathway toward high-throughput CT-metrology without necessitating full CT reconstruction.

Keywords: Computed tomography (CT), metrology, non-destructive testing (NDT), sparse-view, 3D reconstruction, differentiable rendering, 3D–2D, elastic registration, nonrigid registration, radial basis functions (RBFs), view optimization, CT metrology, high-speed

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1. Introduction

X-ray computed tomography (CT) is widely used for dimensional metrology and non-destructive testing (NDT) of high-value aerospace, additive-manufacturing, and biomedical components, enabling internal inspection and CAD comparison for complex geometries that are inaccessible to tactile or optical methods [1, 2]. Key benefits include full-field 3D coverage, Automated general dimension and tolerance assessments as well as the ability to measure wall-thickness and assess internal cavities relative to CAD, and the potential to resolve texture-level features when sufficient resolution is available [1, 2]. Metrological CT nevertheless faces throughput constraints driven by acquisition time, dose, reconstruction cost, and uncertainty control: high-resolution scans require dense angular sampling, pushing scan times into hours, and systematic artifacts and noise can bias edge localization and dimensional measurements [1, 3]. These pressures have motivated reconstructions that explicitly trade projection count for priors and forward-model fidelity, ranging from plug-and-play denoisers and learned unrolled solvers to physics-based differentiable renderers [4, 5, 6, 3].

When CT acquisitions are limited in angle or number of views (ie: sparse-

view CT), reconstruction becomes ill-posed, producing streaking, blurring, and loss of contrast under classical Filtered back projection (FBP) and even under Total Variation (TV)-regularized iterative schemes [7, 8]. Such artifacts can be problematic for metrology because they distort edges and shift apparent surfaces [1, 2]. Classical mitigations include computationally intensive iterative/MBIR pipelines that model system physics and can suppress sparse-view artifacts at the cost of order of magnitude increases in runtime [8, 9]. Learning-based approaches embed priors either as plug-and-play denoisers or as unrolled optimization updates [4, 5, 6] at the cost of generalization, while differentiable forward models have been used to reduce dimensionality by directly fitting neural radiance field representations of the reconstruction space [3]. CAD-constrained inspection can also bypass some reconstruction failure modes by operating in projection space, e.g., matching measured radiographs to CAD projections for feature inspection and using rigid CAD registration to restrict the reconstruction volume [10, 11]. Recent work reports useful reconstructions from as few as a few tens of projections in cone-beam (CB) CT and industrial settings [3, 12], while projection-space inspection can operate with a small set of optimized angles [13, 10]. While these methods including neural radiance field reconstruction can improve sparse-view reconstruction quality, their ability to obtain metrology-grade accuracy is largely not validated and they remain memory- and compute-intensive – requiring up to two orders of magnitude more time than FBP. Furthermore, full-reconstruction approaches still require segmentation to yield measurement-ready surfaces[3, 12].

3D–2D registration aligns and deforms 3D models (volumes or meshes) so that simulated projections match 2D X-ray images thus reducing the dimensionality of the fitting problem by assuming prior geometry information. Compared to conventional CT, this reduces required views and computation and avoids segmentation by directly outputting a surface, but it assumes topology consistency with the initial mesh. In industrial sparse-view X-ray inspection, much of the existing literature focuses on rigid pose estimation or multi-part alignment [14, 15] mirrored by similar frameworks in the medical space [16, 17]. Extending registration to elastic deformation introduces additional challenges: deformation models such as B-spline free-form deformations [18], coherent point drift [19], or RBF fields can capture spatially varying manufacturing deviations but must be regularized to remain smooth and physically plausible. In particular, topology preservation and fold-over prevention motivate constraints inspired by locally-injective mapping and in-

jective deformation processing [20, 21, 22]. Closest to our setting, Calmettes et al. perform RBF-based elastic registration via 3D virtual image correlation using only a few radiographs for lattice metrology, but are limited from full-scale industrial applicability by low projection resolution and a small number of RBFs, due to the complexity of the elastic fitting problem[23]. Although these frameworks validate elastic registration conceptually, no prior work has achieved metrologically accurate elastic registration as a surrogate for CT reconstruction and segmentation.

This is because solving the elastic registration problem is complicated by the need to optimize a high dimensional space with highly nonlinear image similarity metrics. Differentiable rendering (DR) provides gradients of simulated image projections with respect to 3D parameters, enabling end-to-end optimization of shape, pose, and appearance from 2D images [24]. Mesh rasterization DR frameworks achieve this by softening discrete occlusivity operators [25, 24], whereas Monte-Carlo differentiable path-tracers extend DR functionality to physics-based simulation [26, 27]. For X-ray imaging, differentiable projectors compute (exactly or approximately) the sensitivity of line-integral images to geometry or attenuation, supporting both projection-space registration and reconstruction [28, 3]. Projective spatial transformers provide differentiable parameterizations of 2D/3D projection geometry and have been used for fast multi-scale 2D/3D registration [29, 30], which train a neural network to generate digitally reconstructed radiograms. Such methods are complicated by the need to train the forward projector network simultaneously while solving the registration problem, limiting accuracy and generalization.

Despite substantial progress, sparse-view CT remains challenging for metrology because reconstruction artifacts and noise bias edge localization and surface placement, and learned priors are difficult to validate when dimensional accuracy (rather than visual fidelity) is the primary objective [1, 2]. Reconstruction-centric differentiable methods still solve a high-dimensional volumetric inverse problem and often down-sample detector or volume resolution to meet memory and runtime limits [3, 12], and industrial workflows typically still require segmentation to obtain measurement-ready surfaces [10]. As an alternative, CAD-constrained 3D–2D registration can directly deform a shape prior to match projections using compact non-rigid models (e.g., B-spline FFD, CPD, or RBF fields), reducing the effective parameter space and producing a surface without segmentation [18, 19, 31]. However, metrology-grade validation and scalability in sparse-view industrial regimes

remain limited; for example, few-radiograph CAD-bootstrapped RBF registration via 3D-VIC is constrained by low projection resolution and low deformation DOF [23]. Practical pipelines must therefore balance throughput, robustness under sparse views, and topology-consistent deformation while reporting metrology-relevant accuracy against full-view CT and external measurements [11]. Although fast differentiable radiogram simulation has been demonstrated [32], and been applied to sparse-view CT reconstruction [28, 3, 12, 33], it has not been yet applied it to solve the 2D-3D elastic registration problem. Moreover none of these methods have been evaluated for appropriateness for CT metrology (all significantly down-sample CT projections and are bottle necked by high computational overhead), and furthermore do not solve the segmentation challenge.

This work addresses the throughput gap by using 2D-3D elastic registration as a sparse-view direct-to-segmentation method that operates directly in the CT projection space; it uses a combination of strong priors via the CAD and multi-resolution coarse-to-fine RBF-based deformation field to address the data scarcity of the sparse-view fitting problem. We apply differentiable rendering to efficiently solve the complex 3D–2D elastic registration problem, and search for simple heuristics for view selection. Maintaining a metrological focus, dimensional measurement accuracy using elastic registration is directly bench-marked against full-view CT reconstruction/segmentation and coordinate measuring machine (CMM) measurement. Our objective is to demonstrate fully automatable CT metrology in minutes, and open the door to per-part CT metrology.

2. Methods

Elastic registration can serve as a surrogate for CT reconstruction and segmentation when a geometric prior is available; in industrial microCT this prior is typically the nominal CAD, and manufacturing deviations are often small relative to it. We exploit this setting by fitting the CAD directly to sparse projections: a multi-part rigid registration estimates pose and assembly misalignment, followed by an elastic stage that uses an RBF deformation field for non-rigid deviations. A differentiable projector (soft rasterization) provides gradients of projection-space similarity with respect to deformation parameters, enabling efficient, physics-consistent optimization while avoiding voxel reconstruction and downstream segmentation. Finally a multi-resolution strategy is imposed to improve the quality of the registration and

maximize accuracy.

2.1. Radiogram simulation and differentiable rendering

This subsection describes the generation of differentiable simulated microCT X-ray projections.

2.1.1. Differentiable rendering approach

The method leverages recent differentiable rendering technology that allows the direct analytical computation the of the gradient of an image comparison objective with respect to a 3D model that is input into the imaging engine. Differentiable rasterization uses PyTorch3D [34], which provides an optimized implementation of soft rasterization as introduced by Liu *et al.* [25, 35]; rasterization is preferred for its computational efficiency compared to existing frameworks [32]. The essential challenges faced by differentiable rendering via rasterization are the discrete operations that bin faces determining occlusion and the distribution of occlusivity across multiple pixels and depth levels. Rasterization is softened using a continuous probability density, D_j^i, that each triangle, j, contributes to a pixel, i, illustrated in Figure 1. Triangle contributions are then aggregated across depth to yield a continuous occlusion likelihood per pixel, $\hat{S}(p_i)$.

The efficiency of this method, proposed by Liu et. al. [35][25], is improved by PyTorch3D by restricting evaluation to faces that can locally influence each pixel and by capping the number of candidate face contributions to the local vicinity of the pixel through fixed per-pixel binning (i.e., a bounded depth list). The renderer outputs a differentiable depth-sorted list of triangle distances per pixel. These softened depth values form a 3D tensor of shape $[N_y, N_x, N_{\text{faces,max}}]$, where N_x and N_y denote image dimensions and $N_{\text{faces,max}}$ denotes the maximum number of face intersections retained per pixel, and these tensors are subsequently used to compute simulated X-ray projections.

2.1.2. Forward projection and radiogram simulation

Figure 2 schematically summarizes the forward projection procedure. The method computes X-ray attenuation by estimating the path length traveled by each ray inside solid objects using the rendered depth information. Industrial parts and tooling often comprise multiple components with rigid misalignment, such as turbine-blade cores relative to outer shells. The framework therefore treats sub components independently during multi-part rigid registration (see [36]). To avoid computationally expensive and non-differentiable

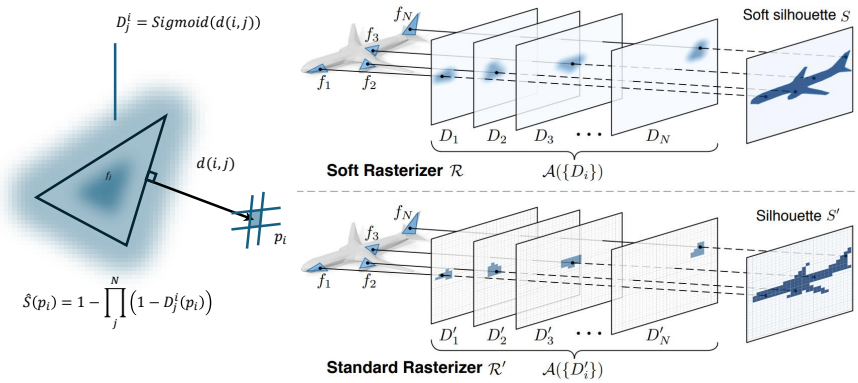

Figure 1: Schematic illustration of the process of differentiable rendering using the soft rasterization technique, implemented in Pytorch3D[34]. Triangle boundaries are 'softened' by a distance-based sigmoidal probability density function that defines triangle/pixel occupancy (Adapted from Liu et al.[25]

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mesh intersection operations, the method applies an optical strategy to emulate boolean merge and subtraction operations on closed volume meshes. To optically simulate mesh merging, ray intersections at regions of mesh overlap are ignored in the distance computation, whereas for a boolean difference, the path length through an interior core (representing a void) is subtracted only when the ray is inside the exterior component. This allows all meshes to be moved and deformed relative to each other while only requiring topological consistency of the initial separate meshes rather than the final mesh.

Two scenes are thus computed for the exterior meshes and the interior meshes, that give two depth map tensors, which encode distances from the source to the faces contributing to each pixel. The optical subtraction of the core meshes from the exterior meshes proceeds by the rule-based merging, sorting, and filtering of the source-triangle intersections given in these two output tensors. Even though the process is discrete and non-differentiable, the differentiability of the source–triangle distances makes the overall process differentiable. The resulting depth maps are transformed to simulate X-ray attenuation and stacked across sampled views to form a differentiable tensor suitable for direct comparison with microCT projections (radiograms). At

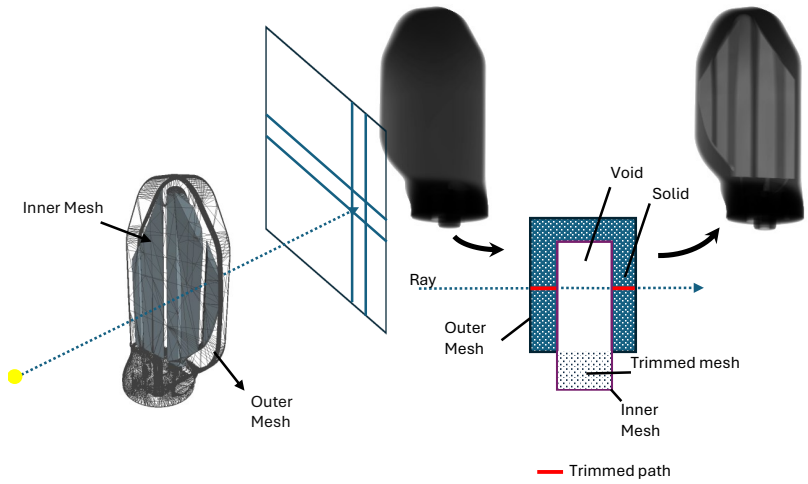

Figure 2: Overview of the Xray forward projection and process of subtracting interior meshes from the exterior meshes. Path distance is computed by subtracting distance traveled through an interior mesh that is inside of an exterior mesh.

present, the simulation omits focal-spot blur and scattering-induced noise, which are reserved for future work,

2.1.3. Image sub-sampling

Image sampling is used to reduce the memory cost associated with differentiable rendering, which can be substantial for large projection images. Two complementary sampling strategies are used: (i) sub-sampling from the set of projections selected for registration (view batching), and (ii) sub-sampling within each projection image (sub-image cropping). In addition, projection images are down-sampled (binned) using a Gaussian pyramid, or interpolation when the binning factor is non-integer; this is done because down-sampling can retain some of the information contained in the original, but substantially reduces memory requirements and practically speaking a binning factor of 2 is sufficient to attain good accuracy.

The projections are separated into batches using a sequential round-robin selection process, which produced the best results compared to stratified and random sampling. Sub-sampling within each image reduces the differentiable rendering domain by rendering only a fixed-size rectangular window. At each iteration, a rectangle of the desired size is extracted from the CT Projections with a random center, and this center is held constant across all images in

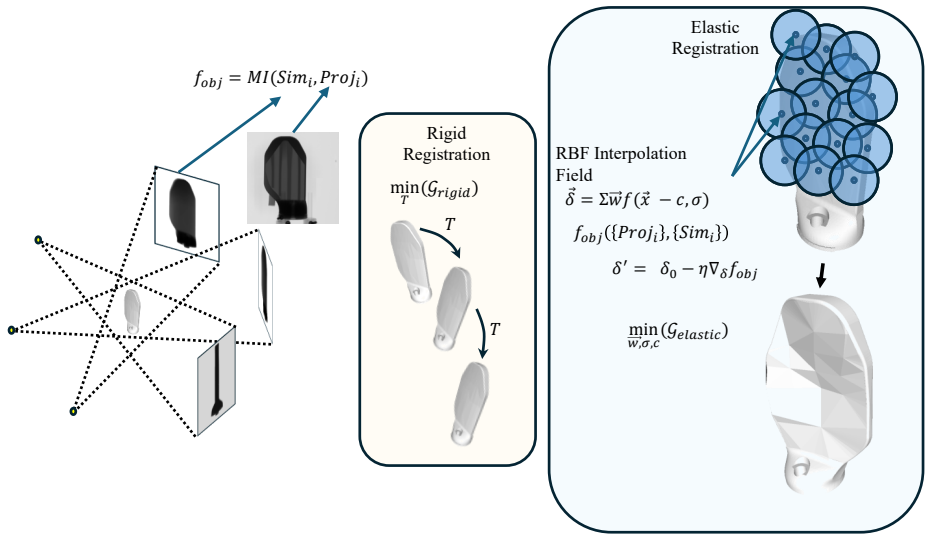

Figure 3: Schematic overview of the proposed registration framework. Simulated and target CT images are compared, and the registration process transforms an initial CAD until the simulated projections match the real projections as closely as possible. Rigid registration determines the translation and rotation matrix, T, and elastic registration determines the deformation field parameters.

the batch so that a consistent region is compared across the batched projections. The rendering region of the simulator is likewise reduced to match the sampled zone of the corresponding image, substantially reducing memory without significantly impacting accuracy. The sub-image size is typically 50% of the original image size in each dimension, up to which the decrease in accuracy was marginal.

2.2. Proposed 2D-3D elastic registration framework

Figure 3 summarizes the registration framework. The method first performs multi-part rigid registration to estimate the orientation and location of each mesh component, and then performs elastic registration to capture manufacturing deviations. Rigid registration is required to remove large alignment errors and to calibrate the simulator so that pose and referential information are computed accurately.

After rigid alignment, elastic registration proceeds in two stages. First, an RBF field represents the deformation field to capture coarse-to-fine geometric variation. Second, the method performs a vertex-level refinement

under explicit smoothness constraints, which is referred to in this work as texture fitting. Although we use relatively coarse meshes in this work which cannot fit surface texture details, the goal is to make a tool that can do that if sufficiently fine meshes were used.

2.2.1. Rigid registration

Rigid registration estimates the rigid transformations that minimize image-comparison metrics between simulated and measured projections. Owing to the noisy and non-convex nature of these metrics, the framework employs gradient-free optimization (see 2.2.6).

$$\min_{R,\mathcal{T}} \sum_{x,y}^{X,Y} \mathcal{G}_{\text{rigid}}(x,y) \tag{1}$$

R and $\mathcal{T}$ denote the rigid rotation and translation parameters, X and Y denote the simulated and measured projection-image sets, (x, y) index images, $\mathcal{G}_{\text{rigid}}(x, y)$ denotes the image-to-image similarity metrics (Section 2.2.3), and $\mathcal{G}_{\text{rigid}}$ denotes the aggregate rigid objective across pixels and views. The method applies rigid registration greedily across components, starting with the outermost components.

2.2.2. Elastic registration

Elastic registration estimates a 3D deformation field $\mathcal{D}$ that minimizes the image-comparison metric across all views.

$$\min_{\mathcal{D}} \sum_{x,y}^{X,Y} \mathcal{G}_{\text{elastic}}(x,y). \tag{2}$$

Here, $\mathcal{D}$ denotes the deformation field applied to the mesh, X and Y denote simulated and measured projection-image sets, (x, y) index pixels, $\mathcal{G}_{\text{elastic}}(x, y)$ denotes the per-pixel elastic similarity contribution (Section 2.2.3), and $\mathcal{G}_{\text{elastic}}$ denotes the total elastic objective.

The method represents $\mathcal{D}$ using sets of radial basis functions. This representation reduces the dimensionality of the optimization and suppresses deformations smaller than the RBF spacing, thereby acting as a low-pass filter that mitigates noise induced by sparse-view sampling. This filtering reduces reliance on explicit constraints that may otherwise bias the solution away from the true optimum. The deformation field evaluated at vertex v_i

is defined using a set of RBF centers c_j for $j \in J$ (where J denotes a local subset of active RBFs).

$$\mathcal{D}(v_i) = \sum_{j \in J} w_j K(v_i) \tag{3}$$

$$K(v_i) = \exp\left(-\frac{\|v_i - c_j\|^2}{\sigma_j^2}\right) \tag{4}$$

where w_j denote RBF weights. In this work Gaussian kernels are used, so that the kernel function K has only the kernel radius σ_j and the kernel center, c_j, as parameters. In practice, the optimization updates w_j, the center locations c_j, and radii σ_j jointly. The method trims the local active set of RBFs, J, by thresholding activation values to avoid evaluating negligible contributions. The weights w_j are initialized to zero, and N RBF centers are sampled on the mesh surface using furthest-point sampling (FPS) to achieve a near-uniform distribution. The base radius σ is set to the mean distance to the five nearest RBF neighbors to ensure mutual interaction and smoothness, and a global radial multiplier adjusts overall smoothness. The method caches and periodically refreshes the active set J because center locations change slowly, improving scaling of memory usage. Two RBF layers (200 and 2000 centers) capture coarse and fine geometric detail.

2.2.3. Image comparison and objective functions

To compare real and simulated projections, we use entropy-normalized mutual information (NMI) for global similarity and pixel-wise energy differences between first- and second-order normalized edge maps. NMI is defined as

$$\mathrm{NMI}(\hat{x}, \hat{y}) = \frac{H(\hat{x}) + H(\hat{y}) - H(\hat{x}, \hat{y})}{H(\hat{x})}, \tag{5}$$

where $H(\cdot)$ is the marginal/joint entropy from histograms and $\hat{x}$ and $\hat{y}$ are the scaled simulated and target images, respectively. Because histogram-based mutual information is typically non-differentiable, we differentiate NMI by softening binning with sigmoid transitions at bin boundaries, using a sparse implementation that accumulates only into N adjacent bins (typically $N = 1$) to reduce memory, with sigmoid scaling chosen so the bin-center value is 0.999.

Edges are extracted using a sigmoid-normalized Sobel operator G.

$$\hat{G}(x) = \mathrm{sigm}(G(x) - \mathrm{mean}(G(x)), \mathrm{std}(G(x))) \tag{6}$$

The sigmoid function is parameterized by the gradient-map mean and standard deviation, yielding a simple and differentiable normalization. Images are scaled via a log transform and normalization, $\hat{x} = f(\log(x))$ with $\hat{x} \in [0, 1]$, where x is clamped to the inverse of the min/max 16-bit integer values. Mutual information is robust for rigid registration because pre-elastic misalignment (warp/scale) can cause imperfect edge correspondence and cause edge-based objectives to skew the location of the true optimum, while elastic registration adds edge penalties to improve accuracy. The per-view objectives are

$$\mathcal{G}_{\text{rigid}}(x, y) = \sum_{x,y}^{X,Y} -\log(\text{NMI}(\hat{x}, \hat{y}))) \tag{7}$$

$$\mathcal{G}_{\text{elastic}}(x, y) = \sum_{x,y}^{X,Y} \begin{pmatrix} -\log(\text{NMI}(\hat{x}, \hat{y})) + 0.6\log(\|\hat{G}(\hat{x}) - \hat{G}(\hat{y})\|) \\ +0.6\log(\|\hat{G}^2(\hat{x}) - \hat{G}^2(\hat{y})\|) \end{pmatrix} \tag{8}$$

2.2.4. Mesh regularization constraints for texture fitting

After RBF fitting, the method performs texture fitting by directly deforming local vertex positions. To suppress noise introduced by stochastic optimization and objective variability, the framework imposes smoothness constraints during texture fitting. Rather than applying classical Laplacian, edge, and normal regularizers, the method penalizes deviation from the RBF-stage output by constraining local curvature, normal orientation, and vertex position relative to the RBF reference mesh. The normalized curvature, $\mathcal{K}$, normal ,$\mathcal{N}$, and position, $\mathcal{P}$, constraints are defined as

$$\mathcal{K} = w_\kappa \frac{\delta_T}{N_p} \sum_i^{N_p} \left| \nabla_s(\hat{n}_i) - \nabla_s(\hat{n}_{i,\text{REF}}) \right|, \tag{9}$$

$$\mathcal{N} = \frac{w_\mathcal{N}}{2N_v} \sum_i^{N_v} \left| (\hat{n}_i) \bullet (\hat{n}_{i,\text{REF}}) \right|, \tag{10}$$

$$\mathcal{P} = \frac{w_\mathcal{P}}{\delta_T N_p} \sum_i^{N_p} \left\| (v_i) - (p_{i,\text{REF}}) \right\|, \tag{11}$$

where $\hat{n}$ denotes normalized face normals, ∇_s denotes the surface-projected gradient, w_κ, $w_\mathcal{N}$, and $w_\mathcal{P}$ denote constraint weights, δ_T denotes mean triangle size, and N_v denotes the number of vertices. The combined constraint

Table 1: Primary elastic registration parameters and nominal settings.

Parameter	Nominal / range	Notes
Number of images (N_{im})	3–30 per case	View angles uniformly sampled or optimized.
View angles	case-dependent	Determined by uniform sampling or view-selection.
Binning factor (BF)	(1-4) typically 2	Down-samples to fit within 16–20 GB GPU memory.
Batch size	50% of samples	Rolling sequential batch sub-sampling of projections.
RBF layers	2	Multi-scale deformation.
Number of RBFs (N_{RBF})	200–500 (coarse); 2000–5000 (fine)	Coarse-to-fine control points for deformation.

term is then

$$\mathcal{C} = w_c(\mathcal{K} + \mathcal{N} + \mathcal{P}) \tag{12}$$

where $\sqsupseteq_c$ is the overall regularization scaling factor, and is added to the objective for the texture fitting step.

$$\mathcal{G}_{\text{texture}}(x, y) = \mathcal{G}_{\text{elastic}}(x, y) + \mathcal{C}. \tag{13}$$

2.2.5. Key elastic registration parameters

Elastic registration performance depends on several parameters. In order of importance to elastic registration accuracy, these are the number of images (N_{im}) and their angles, followed by the binning factor (BF), batch size, and the number of RBFs (N_{RBF}). Views are uniformly sampled by default or chosen by the view-selection method in Section 3.2.4. Projections are processed using rolling sequential batch sub-sampling. Two RBF layers are used to capture multi-scale deformation. Primary parameters and nominal values used in this study are summarized below Table 1. Other relevant parameters include the RBF radius, sub-image sampling method, the learning-rate schedule, and iteration count. The RBF radius is set globally as a multiple of the mean distance to the nearest five RBFs, using 1.0 and 1.25 for the first and second layer, respectively. For sub-image sampling, a random sub-region is extracted from each image, with sub-image size fixed to 0.5 of the original binned image. The learning rate is scale-aware, using an exponential decay

from 10× voxel size to 1/10 of voxel size, and 200 iterations are typically sufficient to achieve good results.

Objective weights are fixed, while texture-fit constraint weights and texture quality/precision are examined. Performance also depends on rigid-registration quality: elastic registration degrades when the initial guess is far from the final shape (approximately 300 pixels), though additional iterations and tuning may extend this range. In the present study following rigid registration, maximum thickness deviation is approximately 25% of the total thickness value, but is typically less than 10 voxels. Rigid registration of an undeformed part is sub-voxel accurate using 4 views [36].

2.2.6. Optimization methods

Rigid registration uses a gradient-free Powell-like solver to handle the noisy, non-convex NMI objective with few parameters. Specifically, the method modifies Powell's line-search procedure by segmenting the gradient vector into orthogonal search directions which are used as the search directions at each iteration (see [36]), reducing the time it takes to do rigid registration while maintaining similar accuracy. Elastic optimization is implemented in PyTorch, using Adagrad to improve robustness under noisy objectives and sparse data. Adagrad updates the parameter vector θ as follows.

$$g_t = \nabla_\theta \mathcal{G}(\theta_t), \tag{14}$$

$$G_t = G_{t-1} + g_t \odot g_t, \tag{15}$$

$$\theta_{t+1} = \theta_t - \eta_t \frac{g_t}{\sqrt{G_t} + \epsilon}, \tag{16}$$

where g_t is the gradient at iteration t, G_t is the element-wise accumulated squared gradients, η_t is the learning rate, $\odot$ denotes element-wise multiplication, and ϵ prevents division by zero. We use an exponentially decaying learning rate, initialized at roughly half the mean triangle spacing (25 μm) and decayed to one-tenth of the unbinned voxel size to match the target precision. Elastic registration typically converges in 200–300 iterations (often ∼100), while texture fitting starts from a well-aligned solution and uses lower learning rates for stability—one-tenth of the RBF-stage initial and final rates.

Image sampling and batch size control memory and computation by evaluating objectives on subsets of views at each iteration. Let $\mathcal{V}$ denote the full

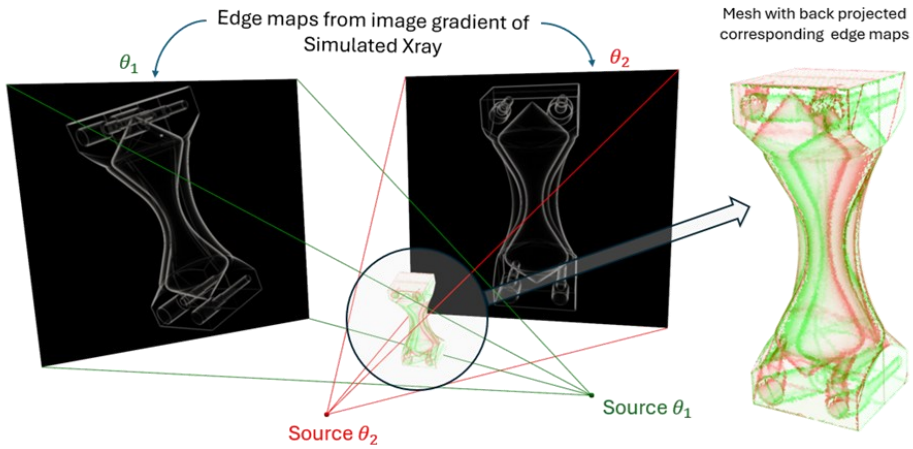

Figure 4: View selection by maximizing uniformity and area of edge map exposure on the actual geometry. See (cf. Singh *et al.*).[37] for comprehensive review of all view angle selection techniques developed and evaluated

view set and let $\mathcal{B}_t \subset \mathcal{V}$ denote the batch at iteration t with $|\mathcal{B}_t| = N_{\text{batch}}$. The generic stochastic update is written as

$$\theta_{t+1} = \theta_t - \eta_t \, \Phi\left(\nabla_\theta \sum_{v \in \mathcal{B}_t} \mathcal{G}_v(\theta_t) \right), \tag{17}$$

where $\mathcal{G}_v$ denotes the per-view objective and $\Phi(\cdot)$ denotes the optimizer-specific scaling (e.g., Adagrad normalization).

2.2.7. View optimization

Several view-optimization strategies are implemented as detailed in our companion publication (see Singh *et al.* [37]). A heuristic is developed for view-optimization, based on the observation that surface positioning accuracy requires adequate representation of the surface in the projection edge maps, accurate registration therefore aims to maximize the uniformity of surface representation, which also helps reduce the effect of the cone-beam artifact occurring due to under-sampling. This technique, called the edge-exposure method (Figure 4), is a purely geometric heuristic that selects views to maximize geometric representation in edge maps.

The edge-exposure objective back-projects simulated edge maps onto the mesh surface. Painted edge intensities saturate each face at a maximum of 1, and face vertices p_f activate a uniformly distributed RBF field with centers c_j for $j \in J$. The objective, $\mathcal{G}_{edge}$ is computed by aggregating face-weighted activation of a gaussian RBF field sampled on the mesh surface.

$$\mathcal{G} = \sum_{j}^{J} \sum_{f}^{\Theta} w_f \exp\left(-\frac{\|p_f - c_j\|^2}{\sigma^2}\right), \tag{18}$$

w_f aggregates edge-map intensities across pixels and views when face f occludes pixel i (i.e., $f \in \Theta(i)$), with X denoting the number of views and N the number of pixels per image.

$$w_f = \text{softmax}\left(\sum_{x=1}^{X} \sum_{i=1}^{N} \begin{Bmatrix} \hat{G}(\hat{x})_i \ \text{ if } f \in \Theta(i), \\ 0 \ \text{ otherwise.} \end{Bmatrix}\right). \tag{19}$$

If gaussian centers are instead set to the measurement points, the edge-exposure technique instead maximizes edge representation in the vicinity of measurement points.

2.3. Test cases and evaluation

The evaluation targets industrial CT metrology and additive-manufacturing QA, with datasets selected to measure wall thickness and accuracy under minimal-view sampling across a range of sizes, morphologies, and surface roughness. Figure 5 visualizes the datasets and measurement protocol. All cases are evaluated against CT reconstruction using filtered back-projection and segmentation using the Dragonfly^{TM} deep-segmentation model (version 2024.2). Two cases are additionally evaluated against CMM measurements to benchmark performance against a metrological gold standard. This subsection summarizes the datasets, measurement definitions, and the experiment matrix.

2.3.1. Datasets used and evaluation framework

Three datasets are selected (i–iii i) with the aim of demonstrating validity across a variety of assembled and complex parts and different materials, and are summarized in Table 2 and Figure 5(a), which lists available data, the number of measurement points (N_δ) used, approximate size, unbinned voxel size, and number of projections N_{proj}. In all cases, measurements are

Table 2: Datasets used for elastic registration. MicroCT using Yxlon FF35 (2860 x 2860 - 0.15 mm pixel pitch)

Dataset	Type	N_δ	Voxel size (0.001")	Size (")	N_{proj}	Description
i: Chinook	MicroCT	14	1.47	3.5	2500	Ti 3D-printed blade - 0°
ii: Art8	MicroCT, CMM	800	1.31	2.5	4500	Ti - 3D printed, polished - 20°
iii: HPT blade	MicroCT	18	1.18	2	3100	Inconel - Cast - Smooth - 30°

evaluated at the measurement points, and benchmark against maximum-quality full-view microCT scans reconstructed using standard FBP. Dataset i) is a titanium 3D-printed root fitting support structure for a Chinook rotor blade with an internal cavity (Figure 5(b)) and relatively high surface roughness. Dataset ii) is an artifact designed for this study that can be split and reassembled to emulate an internal cavity while enabling high-accuracy CMM measurements at predefined locations. The artifact is printed in titanium; both halves are measured by CMM, assembled and scanned via microCT, chemically polished, then re-measured and re-scanned to assess surface-roughness effects. The datasets and precise CMM data acquisition procedures are presented in [38]. Dataset iii) is a cast Inconel high-pressure gas turbine (HPT) blade with a complex interior cavity provided by a gas turbine manufacturer. Key QA objectives include limiting warping, ensuring correct feature placement.

Accuracy is quantified via thickness measurements δ_t at specified points for all cases, computed as the distance between the first intersections of a measurement ray with the inner and outer meshes (Figure 5(b)). Thickness measurement is chosen to illustrate a common measurement desire from CT metrology, during QA and is notably a double-sided measurement and more stringent than simple surface deviations. Mesh-mesh comparisons for our method for this dataset are given in [37].

For Art8 (ii), ray origin and direction are defined from paired inside/outside CMM points measured on each half along a grid, before and after chemical polishing. For the aero-engine component cases (i and iii), the manufacturer specifies measurement points on a surface grid; CAD-defined points are trans-

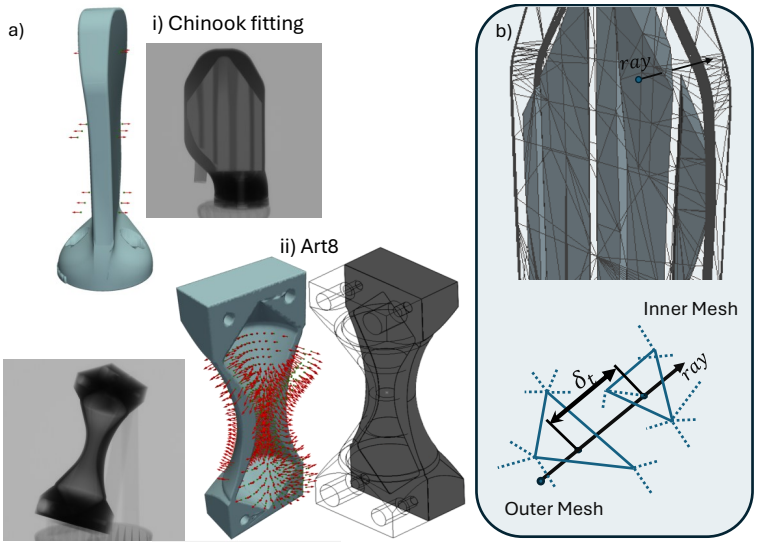

Figure 5: Visualization of datasets and measurement point evaluation. a) shows the geometry, radiogram and measurement point locations of the Chinook fitting (i) and the Art8 (ii). The (iii) HPT blade geometry is not visualized for confidentiality reasons. b) shows thickness measurement computation as the distance between ray intersections with exterior and interior meshes

formed to the CT or registration frame via rigid registration. Because CAD differs from the manufactured geometry, these transformed points may not coincide with the intended physical locations, introducing measurement-point placement error in cases i and iii.

For case ii, CMM points define the measurement locations and must be mapped into the CT and/or CAD frames. Datum-based alignment produced spatially varying placement error (small near alignment features and larger farther away), so we instead align CMM points to the CT reconstruction frame using iterative closest point (ICP) by minimizing point-to-surface distance to the CT reconstruction segmentation mesh, then transform the aligned points to the CAD frame. This minimizes alignment error relative to CT reconstruction, not necessarily relative to elastic registration.

MicroCT scans are acquired on a Comet Yxlon FF35 (2860×2860 detector, 0.15 mm voxel size) and CMM measurements on a GLOBAL S Green 7.10.7 Hexagon CMM at École de Technologie Supérieure (ETS). Elastic registration uses a subset of the projections acquired for CT reconstruction.

Table 3: List and descriptions of test cases executed and datasets involved.

Study	Dataset	Varied	Description
A	i - iii	N_{im}, BF	Evaluate registration accuracy with RBFs
B	i - iii	N_{im}, w_c	Evaluate effect of texture fitting
C	i	N_{RBF}, N_{batch}, R_a	Evaluate parameter sensitivity
D	i,iii	N_{im}, Projections Y	Demonstrate effect of view optimization

2.3.2. Test matrix

Table 3 summarizes the experimental studies. Thickness measurements δ_t are evaluated for all cases. Here, N_{im} is the number of images, BF the binning (down-sampling) factor, N_{RBF} the number of RBFs, N_{batch} the number of images per optimization update, R_a the average surface roughness, and Y the set of images used for registration.

Parameter sweeps vary one parameter at a time about a nominal baseline to limit compute. Each data point is averaged over four independent runs (four registrations or four view optimizations) to reduce stochastic run-to-run variation, which is generally small. The experiments aim to (1) demonstrate CT-reconstruction-like accuracy using elastic registration with far fewer projections than a full CT scan, (2) maximize elastic-registration accuracy and benchmark both elastic registration and CT reconstruction/segmentation against metrological gold standards such as CMM, and (3) minimize the number of views needed for high-quality results. Studies A–C address (1) and (2), while Study D targets (3): Studies A and B evaluate accuracy across all datasets with and without texture fitting, reporting mean absolute thickness deviations and comparing to full-view microCT reconstruction/segmentation (and to CMM for Case ii), Study C performs a parameter ablation, and Study D evaluates maximum accuracy with minimum views possible with view optimization on asymmetric geometries.

3. Results

The proposed method is validated for industrial CT metrology by testing whether elastic registration achieves (FBP) CT-segmentation-like accuracy

with substantially reduced compute and image acquisition time. This section reports accuracy relative to CT reconstruction and segmentation, evaluates accuracy improvements from view optimization, analyzes parameter sensitivities, and summarizes performance considerations.

3.1. Qualitative evaluation of elastic registration

Figures 6 overlays ER surfaces with CT segmentation and edge renderings, showing overall agreement with image edges and CT segmentation on average. Features absent from the nominal CAD can bias registration; for example, residual 3D-print support material in Chinook reduces edge definition and shifts the registered boundary away from the true blade edge. ER nonetheless performs well where CT reconstruction yields blurred boundaries due to myriad artifacts that are difficult to segment (e.g., near the Chinook fitting tip). ER remains uniform, although without CMM it is unclear which is more accurate. ER is limited for small features—cooling-channel ribs are not fully captured when below the RBF spacing—and though texture fitting increases local detail but is constrained by curvature regularization, whose influence grows as feature size decreases and can suppress overly small features (readily alleviated with more refined meshes). Overall, ER achieves accuracy comparable to CT reconstruction and segmentation.

3.2. Elastic registration accuracy

Accuracy is evaluated using thickness measurements and compared with CT reconstruction and segmentation and benchmarked against CMM where available. Because accuracy depends on projection count, the analysis focuses on the number of projections as well as binning factor, view selection, and refinement strategy.

3.2.1. ER accuracy compared to microCT segmentation and effect of number of images and binning factor

Three geometries with interior cavities are evaluated. Figure 7 reports elastic-registration (ER) accuracy as the mean absolute deviation between ER- and CT-reconstruction and segmentation-based thickness measurements, normalized by un-binned voxel size as a function of projection count. sub-voxel accuracy is achieved for the Art8 case, whereas the more complex Chinook fitting and HPT blade cases are generally limited to pixel-level agreement due to increased geometric complexity, higher surface roughness

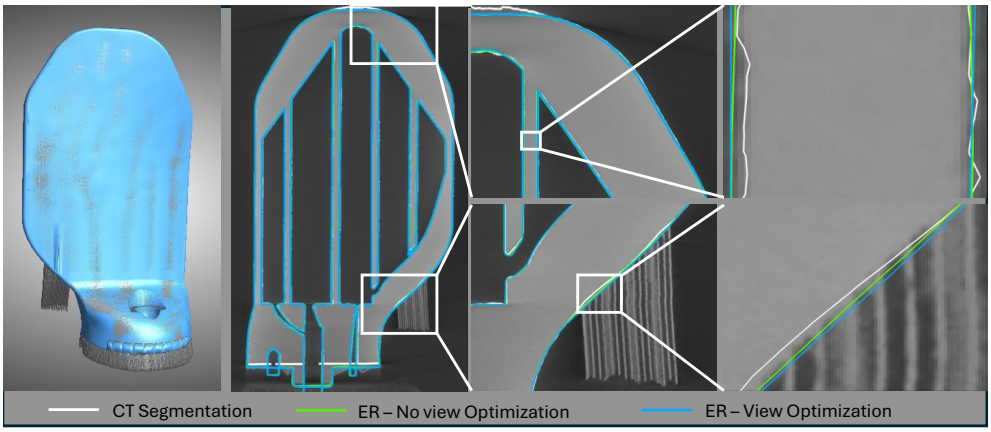

Figure 6: Visualization of the Chinook fitting (i) CT reconstruction volume (left) overlayed onto elastic registration mesh, and cutting plane intersections through Reconstruction volume with edge overlays of Segmentation mesh (white) and elastic registration (ER) meshes with (blue) and without (green) view optimization.

(Chinook), and higher material density (HPT blade). As expected, accuracy improves with additional projections: pixel-level accuracy is typically reached with fewer than 20 images, and sub-voxel accuracy is achieved for the simpler Art8 geometry with as few as 12 projections. Error bars indicate the standard deviation of thickness measurements across the runs, which likewise decreases with projection count and remains a fraction of the mean deviation.

Figure 8 further examines the HPT blade by reporting ER–CT thickness deviation versus binning factor (BF) and projection count. Peak accuracy is on the order of the voxel size, with standard deviation remaining a fraction of the mean error. Full resolution ($BF = 1$) yields the best agreement, reaching pixel-level accuracy with 12 images. When deviations are normalized by the effective (binned) voxel size, agreement improves with increasing BF, indicating that down-sampling preserves sufficient information for pixel-level consistency relative to CT; the maximum ER–CT deviation (over all runs) is approximately three pixels. Accuracy increases with projection count, most strongly at lower BF, whereas higher BF exhibits diminishing returns with

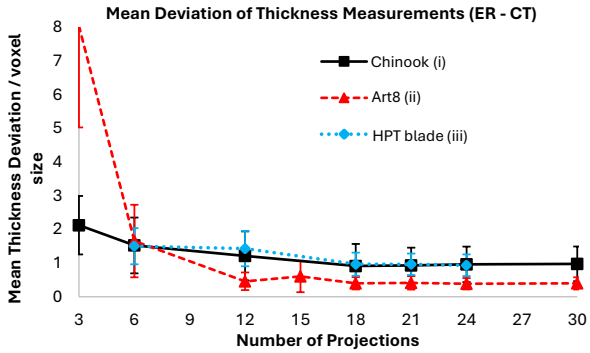

Figure 7: Overall best accuracy of RBF-based elastic registration for all cases (reported as mean absolute thickness deviation measured by ER - CT (FBP) segmentation) normalized by voxel size, versus number of projections. Error bars denote standard deviation of measurements. BF = 2 for all cases

additional views. Voxel-level accuracy is achieved at 12 images for $BF = 1$ and at roughly 18–21 images for $BF = 2$, compared to 3100 CT projections ($\sim$150$\times$ fewer). For the HPT blade pressure-side surface, uncertainty in the CT edge location limits attainable ER–CT agreement. Although $BF = 1$ performs best, runtime and memory increase substantially; in practice, $BF = 2$ provides adequate results.

3.2.2. Effect of texture fitting and comparison with CMM measurements

A constrained vertex-level refinement (texture fitting) after RBF fitting improves accuracy. Figure 9 visualizes the effect of texture fitting (after ER with RBFs) under three constraint weights: low weights propagate noise (particularly in zones where the image-gradient maps become highly noisy), higher weights suppress noise and smooth surfaces, and overly strong constraints reduce the ability to fit texture-level effects; sparse-view artifacts can occur with low constraint weights with 6 views (e.g., vertical surface banding at $w_c = 0.1$, $N = 6$).

Figure 10(a) shows that texture fitting alone performs comparably to RBF-at higher view counts and may improve marginally as views increase; however, RBF fitting improves sparse-view performance, and the combined RBF and texture pipeline yields the best accuracy overall. Figure 10(b) reports thickness accuracy against polished Art8 CMM measurements versus constraint weight: accurate texture fitting requires sufficient constraint for noise damping without over constraining the to prevent fitting small features.

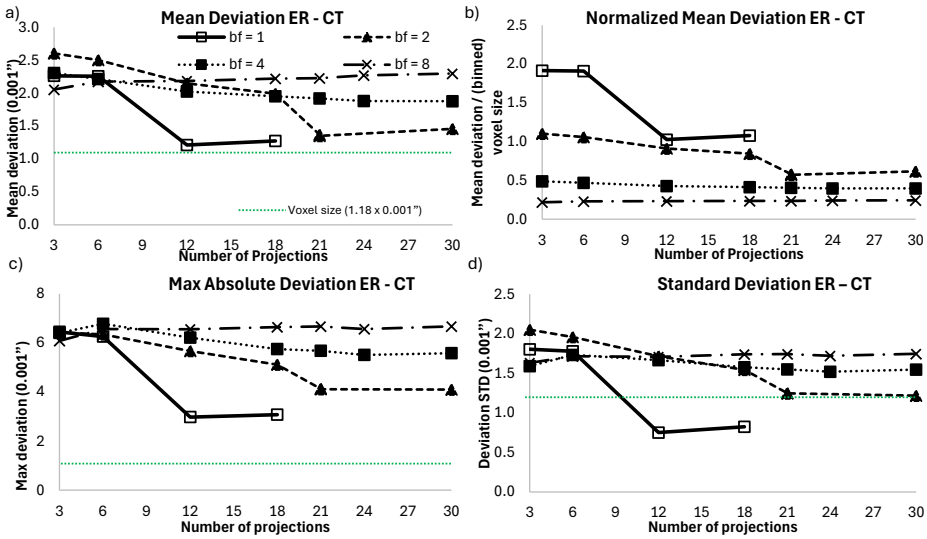

Figure 8: Deviation of thickness measurement HPT blade (iii) (Elastic registration - full-view CT segmentation) versus number of projections and binning factor (BF), a) mean absolute deviation b) mean absolute deviation normalized by effective voxel size c) maximum absolute deviation and d) standard deviation of absolute deviation. Each data point is an average of 4 registrations

Among tested values $w_c = 1$ yields the lowest error (note that constraints are by default normalized by triangle mesh scale and w_c is multiplied by individual constraints, $(w_{\mathcal{K}}, w_{\mathcal{N}}, w_{\mathcal{P}}) = (0.2, 0.5, 0.01)$, optimal tuning is expected to be case-dependent).

Without texture fitting, thickness accuracy remains slightly above voxel size; texture fitting improves agreement and achieves sub-voxel accuracy relative to CT segmentation in the Art8 and HPT blade cases. Relative to CMM, ER attains errors on the order of pixel/voxel size and approaches CT-to-CMM accuracy using as few as 12 views. Peak agreement relative to CT segmentation is higher (approximately half a pixel), while ER-to-CMM accuracy is lower (approximately the voxel size) and comparable to CT-to-CMM accuracy, indicating a practical voxel/pixel-size accuracy floor for optical measurement systems under real projection quality in the absence of denoising or CT reconstruction strategies.

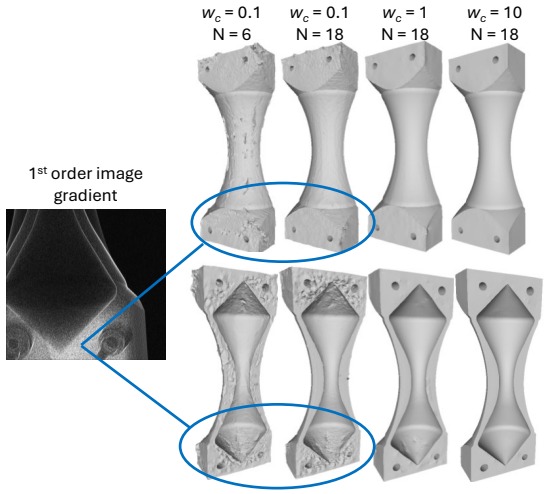

Figure 9: Visualization of texture fitting for three constraint weights on Art8 (without RBF fitting). Edge maps amplify noise which is manifested as poor accuracy in low-constraint case. Vertical streaks appear on low constraint case when $N_{im} = 6$.

3.2.3. Sensitivity of measurement accuracy to registration parameters and performance considerations

While binning factor and number of images impact accuracy most, a parameter study further identifies RBF count and batch size (relative to image-set size) as significant controls on RBF registration accuracy (Figure 11). Sensitivity is case-dependent because RBF count sets the representable geometric detail, while batch size governs memory use and the quality of the update direction per optimization step. Coarse and fine surface finishes are included to capture projection-image quality effects (and the resulting CT reconstruction quality). Other parameters (e.g., RBF radius) are held fixed because their effect is minimal and selection and computation are straightforward 2.2.5.

Figure 11(a) shows that batch size, relative to the total projections, affects accuracy: for a fixed batch size, adding projections generally improves accuracy, but the batch-size fraction is nearly as consequential as projection count. Batch size therefore governs both computation (GPU memory) and accuracy (information per update). Batch-size sensitivity decreases for unpolished Art8, consistent with image-quality-limited performance. Accuracy improves as batch size increases up to ∼50% of the view set, after which gains

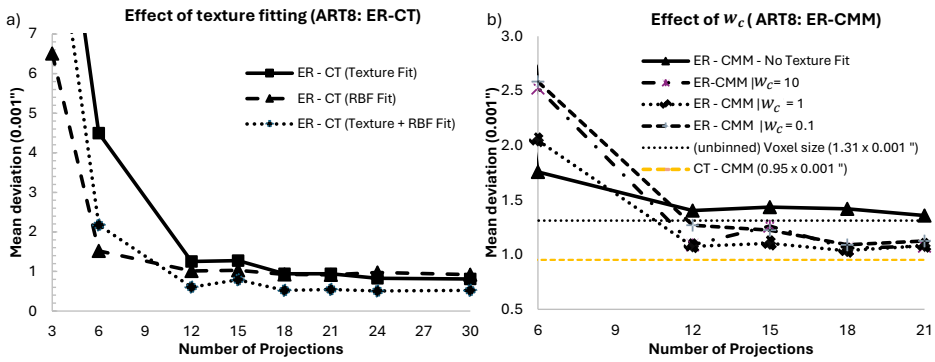

Figure 10: Effect of texture fitting and constraint weight on Art8 thickness accuracy relative to CT and CMM.

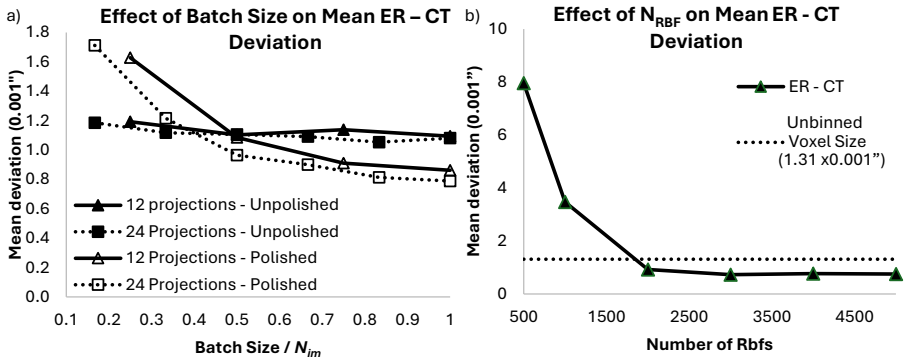

Figure 11: Art8 thickness deviation (ER - Full view CT segmentation) vs (a) batch size and surface finish and (b) number of RBFs and surface finish. $N_{im} = 18$

diminish. A similar trend was found for sub-image sampling (but is not presented here): cropping and random sub-sampling up to ∼50% per dimension yields minimal accuracy loss while reducing memory, so the nominal configuration sets both batch size and sub-image size to ∼50%. Figure 11(b) shows diminishing returns beyond ∼2000 RBFs for Art8; although case-dependent, 2000–5000 RBFs are sufficient in practice for complex cases.

Comparing ER to CT on Art8 before and after chemical polishing shows that reduced surface roughness substantially improves accuracy from voxel-level to sub-voxel and restores a stronger dependence on projection count. In the rough case, increasing image count does not improve accuracy beyond the voxel limit, indicating that image quality dominates in that regime. However, the high roughness of Art8 also caused strong mechanical filtering, precluding

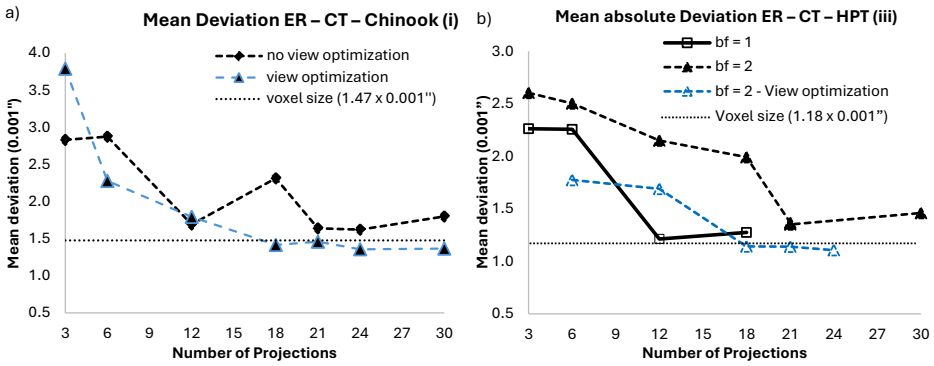

Figure 12: Effect of edge-exposure view optimization on accuracy versus number of images for (a) Chinook (i) (BF = 2) and (b) HPT blade (iii). Blue dashed lines indicate view optimization cases showing view optimization improves accuracy to voxel size and lowers views required for accuracy plateau.

a clean comparison among CT, ER, and CMM measurements.

3.2.4. Effect of view optimization

View optimization improves accuracy for complex, asymmetric geometries with high aspect ratios. While simple cases show marginal gains, view optimization improves consistency and increases accuracy at low projection counts. Figure 12 reports edge-exposure view optimization for the HPT blade and Chinook fitting. For Chinook, view optimization improves consistency and reduces error to slightly below voxel size whereas for HPT blade, the improvement is larger, with $BF = 2$ plus view optimization approaching $BF = 1$ without view optimization (both below voxel size) with reduced view counts. The relative effect of view optimization is visualized in Figures 6 and **??**. The benefit decreases as views increase, but remains significant in sparse-view regimes, enabling voxel-level accuracy with as few as 18 images and improving $BF = 2$ performance to approximately the $BF = 1$ baseline with fewer projections than the non-optimized $BF = 2$ case. The accuracy trend for uniform sampling is also not smooth as images increase because uniform samples can "step over" critical features, whereas it is smoother for view-optimized cases; for a full analysis, the reader is referred to [37].

3.3. Memory usage and computation time

Registration time for RBF fitting and texture fitting scales approximately linearly with memory while the registration can be stored entirely on the 16

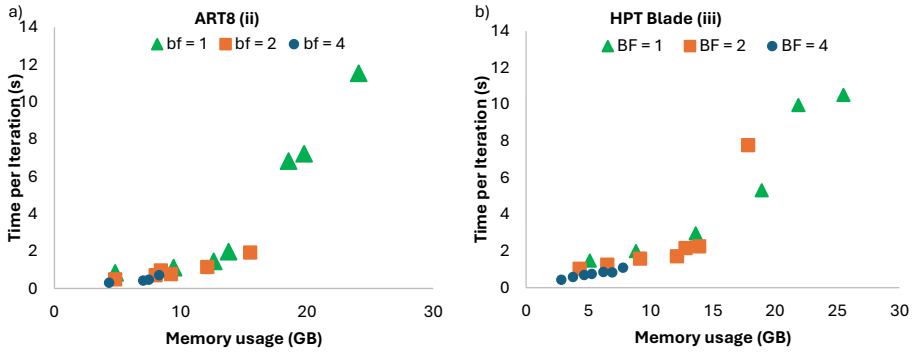

Figure 13: Iteration time versus memory usage for (a) Art8 and (b) HPT blade on an RTX 4070 Ti Super GPU. Compute time - memory scaling is linear below 16 GB limit. Successive points correspond to batch sizes of 3,6,8,11,13,15 images for BF = 2 and 4, and batch sizes of 2,3,4,5,6,7 for BF = 1

GB GPU memory (Figure 13). The dominant cost is differentiable X-ray rendering, which tracks intersections over up to 36 depth layers in this work; image size and mesh size can therefore strongly affect computational cost. Nevertheless, while memory remains below on-board GPU capacity, scaling is approximately linear and iterations typically take < 2 s even for BF = 1. Full-accuracy production use therefore benefits from adequate GPU resources, although the present work is executed on a single consumer-grade GPU and achieves high accuracy. Multi-GPU parallelization is not implemented, but could be achieved by distributing batched images and gradient averaging in a straight-forward manner.

Table 4 indicates mesh-size dependency: Art8 runs approximately twice as fast as the HPT blade and Chinook fitting. Although Chinook has fewer triangles than Art8, compute time is greater for $BF = 2$ despite lower memory usage, due to overhead from optically subtracting the core mesh from the exterior mesh; when internal geometries are present, internal and external meshes are rendered separately, effectively doubling compute. A mesh-size dependence remains, however and compute time scales linearly with mesh size while utilizing less than the GPU memory. With the current configuration, the largest mesh that can be handled with acceptable accuracy and registration time ($<$ 15 minutes) has approximately 4×10^5 faces, which remains a bottleneck for production-scale environments requiring finer meshes to capture texture-level, voxel-sized defects. Despite per-iteration times reaching up to 4 s for full-sized images, registration typically converges within 100–

Table 4: Compute time per iteration per image and memory used per image at different resolutions on RTX 4070 Ti Super GPU

Case	$N_{Triangles}$	BF = 1	BF = 2	BF = 4
i:	98,790	0.345 s | 3.51 GB	0.157 s | 1.25 GB	0.0330 s | 0.237 GB
ii:	78,344	0.372 s | 3.01 GB	0.0805 s | 0.812 GB	0.0033 s | 0.294 GB
iii:	248,350	0.745 s | 4.18 GB	0.144 s | 1.13 GB	0.0521 s | 0.428 GB

Table 5: best mean and std thickness measurement deviation between CT and ER, and acquisition time comparison for three datasets. Deviations are given as a fraction of the un-binned voxel size

Dataset	$\|\Delta\delta_t\|_{avg}$	$\|\Delta\delta_t\|_{std}$	t_{ER} [1]	$t_{ER,Scan}$ [2]	$t_{CT,Scan}$
i: Chinook (CT)	0.9226	0.735	8 m 4 s	38 s	1 h 36 m
ii: Art8 (CT)	0.398	0.323	5 m 20 s	22 s	1 h 58 m
ii: Art8 (CMM)	0.813	0.627	5 m 20 s	22 s	1 h 58 m
iii: HPT blade (CT)	0.951	0.641	7 m 40 s	52 s	3 h 46 m

200 iterations, total elastic registration time is generally below ∼5 min on our configuration with acceptable accuracy; this is comparable to the fastest CT reconstruction and can be further reduced with appropriate compute hardware investment.

3.4. Summary

Table 5 summarizes the best results for each dataset along with total registration time. Art8 achieves sub-voxel accuracy relative to CT segmentation and near voxel-level accuracy relative to CMM. The Chinook fitting and HPT blade are more challenging due to lower image quality from high surface roughness (Chinook) and high-density material (HPT blade), giving mean absolute deviations relative to CT on the order of the voxel size. Image quality strongly limits achievable accuracy: these results use real projections, whereas clean simulated images (even with McRay scattering) yielded errors as low as 0.1 voxels. Overall, CT-reconstruction-like accuracy is achieved with fewer than 20 images in all cases, reducing scan and processing time by approximately 150× (Chinook) to 260× (HPT blade) while maintaining accuracy similar to the voxel size.

4. Discussion

4.1. Accuracy and application to Industrial CT metrology

Measurement of wall thickness using ER agrees with CT segmentation measurements within half a voxel on average for Art8, while more challenging cases (Chinook and HPT blade) reach roughly voxel-level agreement due to lower image quality from high surface roughness (Chinook) and high-density materials with complex shapes (HPT blade); overall accuracy is governed by image quality and geometric complexity. While ER can closely match CT reconstruction/segmentation and obtain sub-voxel accuracy, CMM comparisons indicate that practical accuracy is effectively limited to approximately the voxel size. Additional experiments with ER using simulated images do find sub-voxel measurement accuracy to scale with image noise, which is consistent with prior findings [23]. Also of note, as binning or subsampling can retain metrological information, sub-voxel accuracy must be evaluated relative to the unbinned voxel size.

It remains unclear whether CT can reliably support sub-voxel accuracy and texture-level fitting of surface roughness. First- and second-order Sobel terms combined with mutual information are critical for edge placement and surface localization, but they amplify even modest noise, making distortion-free denoising essential. Several denoising approaches were evaluated, including neural edge extraction and wavelet disentanglement, but these are deferred to future work.

Multi-scale RBF fitting, texture fitting, and view optimization together bridge the sparse-view information gap by imposing strong priors and acting as a low-pass filter against noise and typical sparse-view reconstruction defects. The method begins over-constrained and progressively relaxes constraints: an initial RBF field limits deformation to the RBF spacing to prevent artifacts without skewing the objective, unlike classical edge/normal/curvature regularization, enabling high registration accuracy. Finer detail is then introduced by increasing the number of RBFs and applying texture fitting. Small-feature recovery remains limited to deformations on the order of the RBF spacing (e.g., cooling-channel ribs), and although texture fitting improves local detail, it is still moderated by regularization; constraint weights therefore trade off noise suppression against fine-detail capture and accuracy.

[1]Elastic registration time only; rigid registration is assumed to be done once only.
[2]Not including the CT angular travel time.

4.2. Automation and implications for industrial QA workflows

The method achieves accuracy comparable to full-view CT reconstruction, including when benchmarked against CMM, while requiring far fewer images and offering strong potential for automation. sparse-view reconstruction defects are eliminated, and the pipeline is reliable and fast for metrology. Because radiogram acquisition dominates CT metrology time, scan and processing time are theoretically reduced (excluding travel time) to as low as 22s and less than a minute in all cases, a hundred-fold improvement relative to full microCT for very similar accuracy.

Using the initial geometry as a prior also implicitly performs segmentation, simplifying downstream metrology and supporting high-throughput workflows. For example, producing a high-quality surface segmentation of the HPT blade required several hours of expert manual work with advanced ML segmentation tools, whereas a comparable surface is produced directly by the proposed approach. Finally, operating in projection space yields clearer part edges and is less affected by scattering-induced ambiguity in highly convex parts (e.g., the pressure side of turbine blades), and can therefore outperform CT reconstruction/segmentation in surface quality in some cases.

4.3. sparse-view behavior and view optimization

Accuracy increases with projection count but reaches a plateau at with increased number of projections; under uniform sampling, accuracy gains typically saturate at $\sim$30 images. View optimization uses a fast, purely geometric heuristic from the rigid CAD fit, running in less time than elastic registration. Edge-exposure selection improves accuracy and reduces required views for both the HPT blade and Chinook fitting, with the largest gains with complex geometric cases. For details, see Singh *et al.* [37].

4.4. Sensitivity to key parameters

Dominant parameters affecting accuracy are the number of images, binning factor, batch size, and number of RBFs. Batch size is nearly as consequential as the number of images, with accuracy improving up to $\sim$50% of the selected view set and diminishing returns beyond. Indeed, batch size can be viewed as the key parameter: Figure 11 shows that doubling the number of images while keeping the same batch-size ratio yields minimal gains in accuracy. Because batch size is also the determining factor for memory usage, it effectively governs the tradeoff between compute cost and accuracy.

Sub-image sampling behaves similarly; cropping up to ∼50% per dimension introduces minimal accuracy loss while substantially reducing memory.

4.5. Computational performance and scalability

Runtime scales approximately linearly with memory usage while below GPU capacity, and typical iterations remain < 3 s even for un-binned images. Differentiable rendering dominates compute cost, including depth-layer tracking and per-pixel face interactions, and mesh size therefore strongly affects runtime. Under the current configuration, the practical upper mesh size for adequate accuracy is approximately 4×10^5 faces on a single consumer-grade GPU, though increasing GPU memory reduces this bottleneck. Multi-GPU parallelization is not implemented, but could be achieved relatively simply by distributing batched images across multiple GPUs. In general it was also found that using commercial or industrial GPUs with higher memory effectively removed any bottlenecks to image and mesh size.

4.6. Scope, assumptions, and limitations

Topological consistency of the output mesh prevents detection and visualization of cracks and other defects not represented in the CAD topology, and it is currently unclear if surface level texture details such as an engraving on the surface can be accurately reconstructed. Material density is not explicitly modeled; multi-part or multi-material parts are therefore not considered at this time and are left for future work. The projection model also omits focal-spot blur and scattering noise. Measurement-point alignment remains a source of error, particularly when points are defined in CAD space and transferred to the manufactured geometry. Current constraints during texture fitting may over-constrain small features, motivating improved multiscale or adaptive constraint strategies. Mesh size is currently limited, and substantially more refined meshes are required to capture and evaluate real texture detail.

4.7. Future work

Future work will incorporate more realistic physics-based forward models (focal-spot blur, scatter, noise), advanced denoising, and robustness studies across acquisition conditions to improve accuracy and competitiveness with state-of-the-art CT reconstruction. Performance and scalability to large meshes will be improved, while small-feature recovery will be enhanced via

adaptive RBF spacing and local mesh refinement during registration. Wavelet-disentangled neural edge extraction and neural style-transfer denoising are being explored to improve performance for noisy CT, and defect detection beyond topology-preserving registration will be pursued using hybrid or implicit surface representations.

Because the registration paradigm differs fundamentally from CT reconstruction, it also enables new directions, including physics-constrained strain-field reconstruction, reconstruction of very large parts from partial radiograms at arbitrary orientations, dynamic 4D reconstruction of moving objects (e.g., cardiovascular CT or intraoperative angiography), and end-to-end QA through integrated simulation. Further automation could also be achieved by integrating visual-LLMs to interpret results and auto-flag problematic components on the assembly line.

5. Conclusion

Our proposed ER framework works as a practical surrogate for CT reconstruction and segmentation for industrial metrology. When an initial geometry estimate is available, the method outputs a segmentation-like surface with measurement thickness accurate to within a voxel of the full microCT segmentation using only between 12-20 projections and with no manual input, providing similar accuracy to full-view microCT using filtered back-projection and segmentation. The proposed projection-space framework leverages CAD priors and a hierarchical multiscale elastic registration strategy—using multiple RBF layers followed by constrained vertex-level displacement to progressively introduce detail at finer scales—thereby addressing data scarcity in sparse-view reconstruction. Compared to CMM, our method achieves nearly identical accuracy as full CT reconstruction, approaching the voxel size. Notably, it was found that as binning or subsampling can retain metrological information, accuracy claims must be evaluated relative to the un-binned voxel size to truly gauge whether sub-voxel accuracy is achieved. Overall, projection acquisition time is reduced to as little as 22 s, with total processing time being less than five minutes on a consumer grade GPU, leading to a hundred-fold reduction in acquisition and processing time compared to conventional microCT metrology. Furthermore, it was found that a simple geometric heuristic-only view-optimization approach was sufficient to substantially reduce the required number of projections for asymmetric complex geometries. The method is however limited in the ability to

capture fine details on the order of the mesh size, and is furthermore presently limited to meshes with 4×10^5 faces.

In summary, across representative industrial datasets, the Elastic registration eliminates reconstruction artifacts and attains accuracy comparable to full-view FBP CT reconstruction – 0.4 voxels for case (ii) (0.00051" or 12.9 μm), while achieving a 150–250× reduction in scan time. These results support automated, high-throughput inspection workflows - enabling CT metrology on a per-part basis.

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7. Author Contributions

Hristo Valtchanov: Conceptualization, methodology, investigation, software, validation, formal analysis, supervision, writing – original draft, visualization. Weiqiang Liu: Investigation, formal analysis, methodology, software, writing - review & editing. Sadaival Singh: software, formal analysis, validation, writing - review & editing, visualization. Josselin Paccoud: data curation. Ammar Al Sheghri: validation. Julien Dompierre: conceptualization, methodology. Catherine Desrosiers: Conceptualization, methodology. Nicolas Piché: conceptualization, resources, funding acquisition, Supervision, Investigation, Vladimir Brailovski: project administration, resources, supervision, funding acquisition, writing - review & editing. François Guibault: Conceptualization, project administration, resources, supervision, funding acquisition, Writing – review & editing,

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